Mystery Meat Analysis

BIOLM0051

Archie Benn

Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

Introduction

CITES was established in 1975 as a key legal framework to protect endangered species from global trade, yet from 1975 to 2014 the volume of CITES-listed organisms traded per year quadrupled to over 100 million whole-organism equivalents, a level which directly threatens both populations and whole ecosystems [1]. Rarity status attributed to a species often increases its price and demand, with critically endangered species often fetching the highest prices [2, 3]. The fallout of this is a positive feedback loop between rarity and demand which may drive an endangered species towards extinction [4].

Given the challenge of identifying meat by morphology alone

Methods

Pipeline Overview

Results

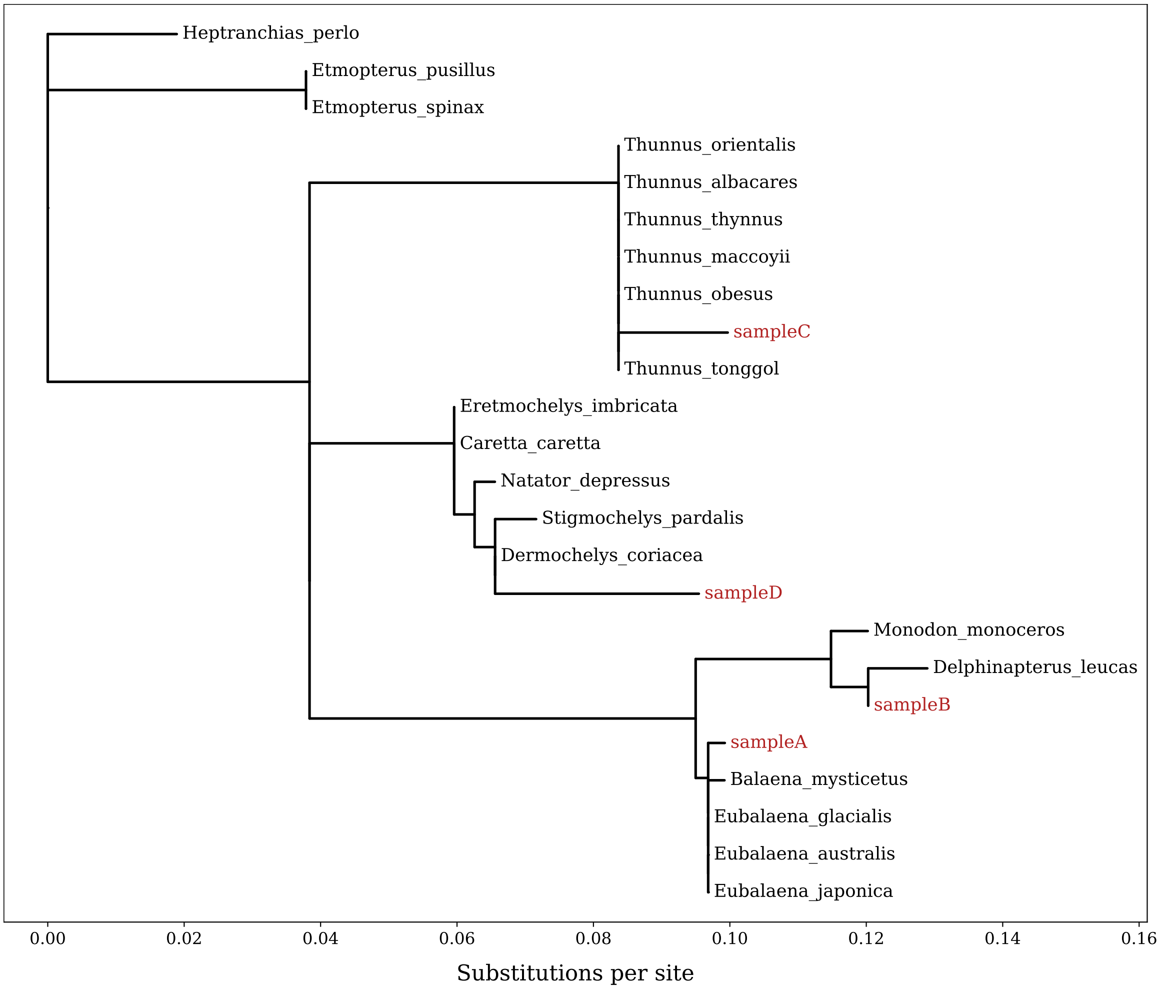

Figure 1: Maximum-likelihood phylogenetic tree with placement of unknown meat samples and their top BLAST hits

Discussion

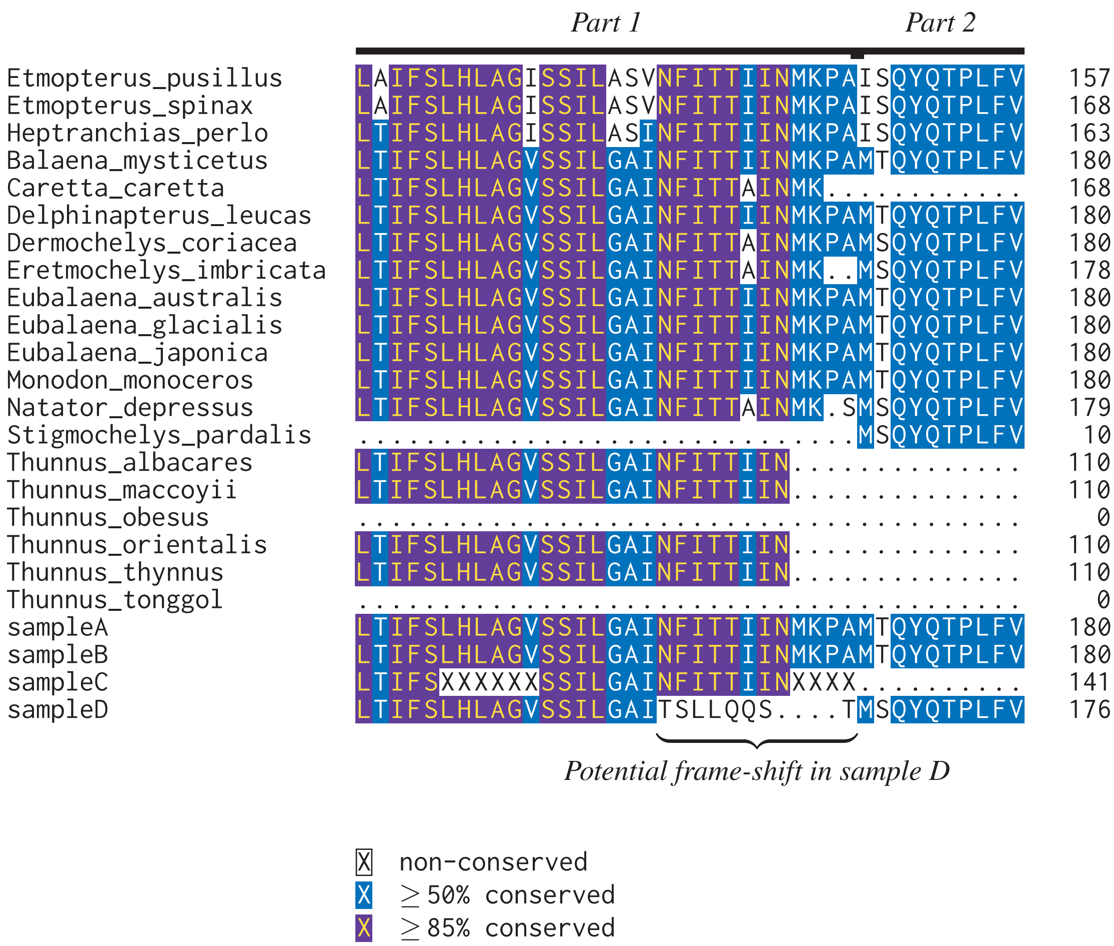

Figure 2: Window of supermatrix alignment with potential frame-shift error in `sampleD` near the highly-conserved `NFITTIIN` region.

```
Dermochelys_c     441  CACCTAGCTGGTGTTTCATCAATTTTAGGAGCTATTAACTTCATTACTAC     490
                       |||||||||||||||||||||||||||||||||||| |||||||||||||
sampleD_part1     451  CACCTAGCTGGTGTTTCATCAATTTTAGGAGCTATT-ACTTCATTACTAC     499
```

Figure 3: Needleman-Wunsch global pairwise alignment at the tail end of `sampleD_part1` against closest reference sequence (*Dermochelys coriacea*). The alignment gap indicates a possible base miss at position 486 in `sampleD_part1` during sequencing.

References

[1] Michael Harfoot, Satu AM Glaser, Derek P Tittensor, Gregory L Britten, Claire McLardy, Kelly Malsch, and Neil D Burgess. Unveiling the patterns and trends in 40 years of global trade in cites-listed wildlife. *Biological Conservation*, 223:47–57, 2018.

[2] Yik-Hei Sung and Jonathan J Fong. Assessing consumer trends and illegal activity by monitoring the online wildlife trade. *Biological conservation*, 227:219–225, 2018.

[3] Franck Courchamp, Elena Angulo, Philippe Rivalan, Richard J Hall, Laetitia Signoret, Leigh Bull, and Yves Meinard. Rarity value and species extinction: the anthropogenic allee effect. *PLoS biology*, 4(12):e415, 2006.

[4] Liam J Hughes, Oscar Morton, Brett R Scheffers, and David P Edwards. The ecological drivers and consequences of wildlife trade. *Biological Reviews*, 98(3):775–791, 2023.

Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Exploratory Figures for Manual Correction of Suspected Sequencing Error

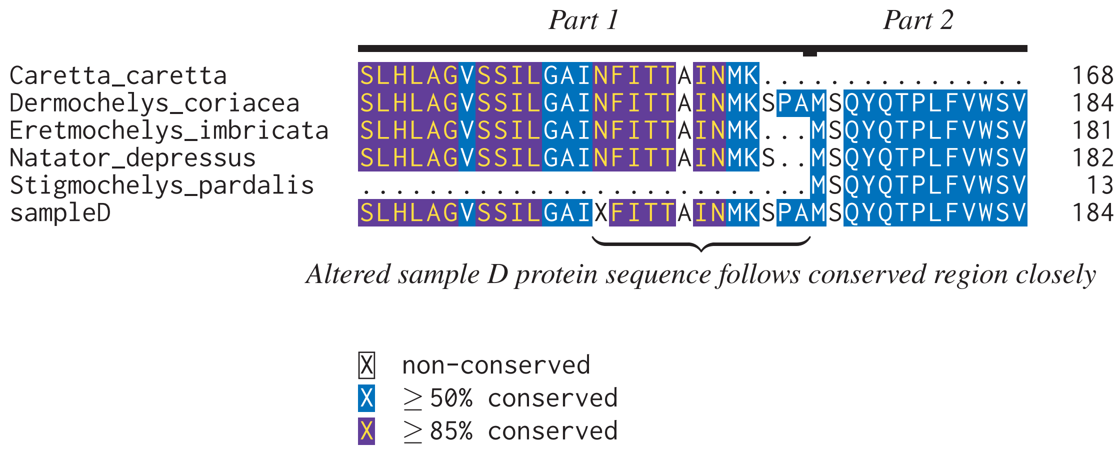

Figure 4: Subset of supermatrix alignment with `sampleD` and innner clade after inserting one ambiguous base (N) to the nucleotide sequence of `sampleD_part1`. This base resolves the apparent frame-shift, resulting in a protein sequence which aligns much more closely to highly-conserved `NFITTIIN` region compared to in Figure 2.

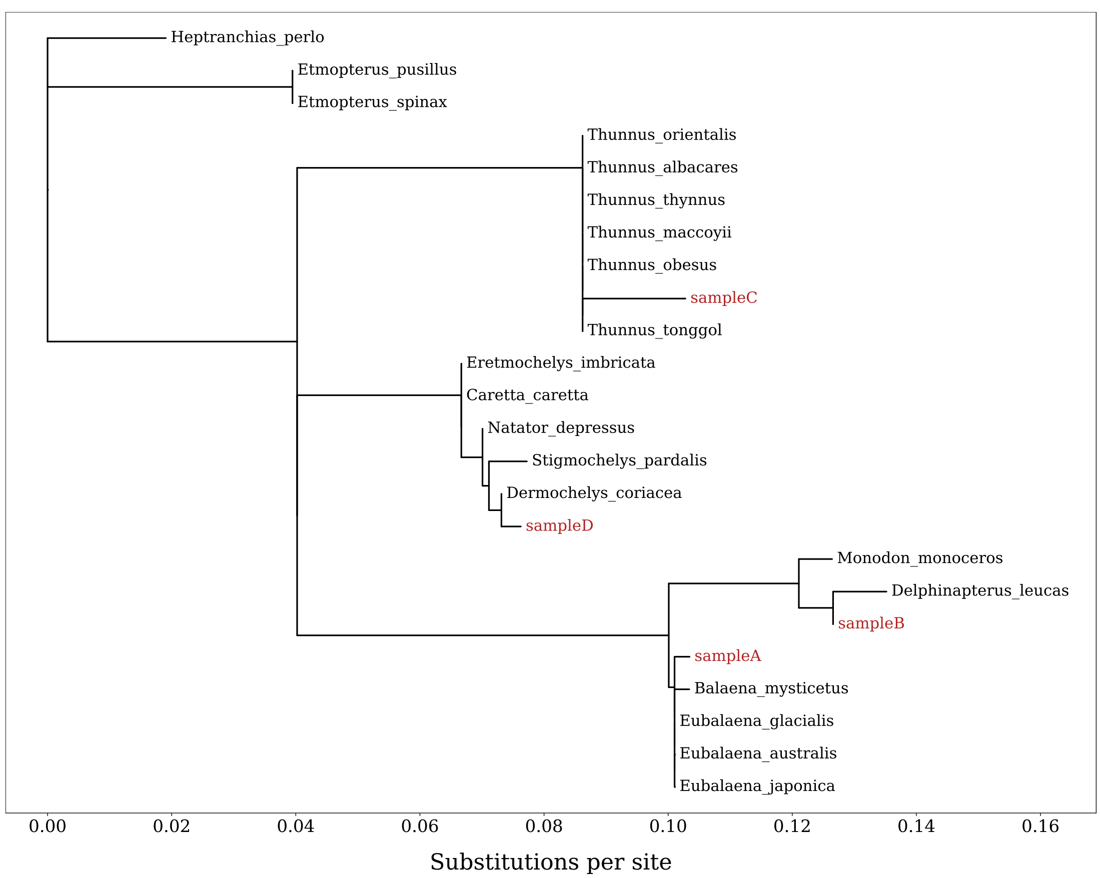

Figure 5: Maximum-likelihood phylogenetic tree after manually corrected alignment in which one ambiguous base (N) was inserted into `sampleD_part1`. This tree is shown to demonstrate the impact of sequencing error and downstream frame-shift on branch lengths compared to Figure 1.